

Surface tension correlations from symbolic regression

$$T_r = \frac{T}{T_c}, \quad F_1 = 1 - T_r$$

Equation 1

Complexity: 1

Loss: 19.9917

$$\sigma = 7.611$$

Equation 2

Complexity: 3

Loss: 0.852208

$$\sigma = (1 - T_r)48.25$$

Equation 3

Complexity: 5

Loss: 0.00143133

$$\sigma = ((1 - T_r)26.19)^{(1.351)}$$

Equation 4

Complexity: 7

Loss: 0.000107575

$$\sigma = (((1 - T_r)29.08) - 0.1882)^{(1.297)}$$

Equation 5

Complexity: 9

Loss: 1.17527e-5

$$\sigma = (((1 - T_r)32) - 0.7083)^{(1.258)} + 0.4981$$

Equation 6

Complexity: 11

Loss: 6.78435e-6

$$\sigma = (((1 - T_r)((0.8331)^{(1 - T_r))27.16)) - 0.08929)^{(1.368)}$$

Equation 7

Complexity: 13

Loss: 3.19751e-6

$$\sigma = (((1 - T_r) + ((1 - T_r) + (1 - T_r)))8.27)^{(((1 - T_r))^(-0.02633))1.341}$$

Equation 8

Complexity: 15

Loss: 1.10972e-7

$$\sigma = (((1 - T_r) + ((1 - T_r) + (1 - T_r)))8.155)^{(((1 - T_r))^(-0.02946))1.345} + 0.02539$$

Equation 9

Complexity: 19
Loss: 6.68887e-8

$$\sigma = (((1 - T_r) + ((1 - T_r) + (1 - T_r))^{((1 - T_r)2.385)^{-0.0596}})) + (1 - T_r))4.322^{(1.277)1.982}$$

Equation 10

Complexity: 21
Loss: 4.98012e-8

$$\sigma = (((1 - T_r) + ((1 - T_r) + ((1 - T_r) + (1 - T_r)))^{((1 - T_r)2.548)^{-0.04066}}))4.302^{(1.296)1.918}) + 0.02232$$